$$k = \left(\frac{1}{t}\right) \ln \left(\frac{[A_0]}{[A]}\right)$$

$$10^{-5} s^{-1} \times 100 \times 60 \text{ s} = \ln \left(\frac{[A_0]}{[A]}\right)$$

$$0.06 = \ln \left(\frac{[A_0]}{[A]}\right)$$

$$\frac{[A_0]}{[A]} = e^{0.06}$$

$$\frac{[A_0]}{[A]} = 1.06$$

$$\therefore \frac{[A_0]-[A]}{[A_0]} \times 100\ \%$$

$$= \left(1-\frac{[A]}{[A_0]}\right) \times 100\ \%$$

$$= \left(1-\frac{1}{1.06}\right) \times 100\ \%$$

$$= 5.6\ \%$$

Example 6

Show that in case of first order reaction, the time required for 99.9% completion is nearly ten times the time required for half completion of the reaction.

Let

$[A_0] = 100$;

when $t = t_{99.9\%}$; $[A] = (100-99.9) = 0.1$

$$k = \frac{2.303}{t} \log \left(\frac{[A_0]}{[A]}\right)$$

$$t_{99.9\%} = \frac{2.303}{k} \log \left(\frac{100}{0.1}\right)$$

$$t_{99.9\%} = \frac{2.303}{k} \log 1000$$

$$t_{99.9\%} = \frac{2.303}{k} (3)$$

$$t_{99.9\%} = \frac{6.909}{k}$$

$$t_{99.9\%} \simeq 10 \times \frac{0.69}{k}$$

$$t_{99.9\%} \simeq 10\ t_{1/2}$$

Evaluate yourself:

(1) In a first order reaction $A \longrightarrow products$ 60% of the given sample of A decomposes in 40 min. what is the half life of the reaction?

(2) The rate constant for a first order reaction is $2.3 \times 10^{-4} s^{-1}$ If the initial concentration of the reactant is 0.01M . What concentration will remain after 1 hour?

(3) Hydrolysis of an ester in an aqueous solution was studied by titrating the liberated carboxylic acid against sodium hydroxide solution. The concentrations of the ester at different time intervals are given below.

Time (min)	0	30	60	90
Ester concentration mol L^{-1}	0.85	0.80	0.754	0.71

Show that, the reaction follows first order kinetics.

7.7 Collision theory :

Collision Theory was proposed independently by Max Trautz in 1916 and William Lewis in 1918. This theory is based on the kinetic theory of gases. According to this theory, chemical reactions occur as a result of collisions between the reacting molecules. Let us understand this theory by considering the following reaction.

$$A_2(g) + B_2(g) \longrightarrow 2AB(g)$$

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